PULLBACKS OF SHEAVES AND STALKS ALONG MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 0034: sources/stacks-chow/tags.txt, source lines 121–121. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03G2: sources/stacks-chow/tags.txt, source lines 4513–4513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00X1: sources/stacks-chow/tags.txt, source lines 1203–1203. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04JC: sources/stacks-chow/tags.txt, source lines 5933–5933. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04D3: sources/stacks-chow/tags.txt, source lines 5713–5713. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05V1: sources/stacks-chow/tags.txt, source lines 7535–7535. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04K0: sources/stacks-chow/tags.txt, source lines 5956–5956. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04K1: sources/stacks-chow/tags.txt, source lines 5957–5957. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 00WU: sources/stacks-chow/tags.txt, source lines 1196–1196. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00VC: sources/stacks-chow/tags.txt, source lines 1142–1142. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 04JZ: sources/stacks-chow/tags.txt, source lines 5955–5955. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S.*

1. *The functor $f_{small}^{-1} : Ab(Y_{étale}) \to Ab(X_{étale})$ is exact.*
2. *The functor $f_{small}^{-1} : Sh(Y_{étale}) \to Sh(X_{étale})$ is exact, i.e., it commutes with finite limits and colimits, see Categories, Definition 0034.*
3. *For any étale morphism $V \to Y$ of algebraic spaces we have $f_{small}^{-1} h_V = h_{X \times_Y V}$.*
4. *Let $\overline{x} \to X$ be a geometric point. Let $\mathcal{G}$ be a sheaf on $Y_{étale}$. Then there is a canonical identification*
$$(f_{small}^{-1}\mathcal{G})_{\overline{x}} = \mathcal{G}_{\overline{y}}.$$
where $\overline{y} = f \circ \overline{x}$.

Proof. Recall that f_{small} is defined via $f_{spaces,small}$ in Lemma 03G2. Parts (1), (2) and (3) are general consequences of the fact that $f_{spaces,étale} : X_{spaces,étale} \to Y_{spaces,étale}$ is a morphism of sites, see Sites, Definition 00X1 for (2), Modules on Sites, Lemma 04JC for (1), and Sites, Lemma 04D3 for (3).

Proof of (4). This statement is a special case of Sites, Lemma 05V1 via Lemma 04K0. We also provide a direct proof. Note that by Lemma 04K1. taking stalks commutes with sheafification. Let $\mathcal{G}'$ be the sheaf on $Y_{spaces,étale}$ whose restriction to $Y_{étale}$ is $\mathcal{G}$. Recall that $f_{spaces,étale}^{-1}\mathcal{G}'$ is the sheaf associated to the presheaf
$$U \longrightarrow \operatorname{colim}_{U \to X \times_Y V} \mathcal{G}'(V),$$
see Sites, Sections 00WU and 00VC. Thus we have
$$\begin{aligned}(f_{spaces,étale}^{-1}\mathcal{G}')_{\overline{x}} &= \operatorname{colim}_{(U,\overline{u})} f_{spaces,étale}^{-1}\mathcal{G}'(U) \\ &= \operatorname{colim}_{(U,\overline{u})} \operatorname{colim}_{a:U \to X \times_Y V} \mathcal{G}'(V) \\ &= \operatorname{colim}_{(V,\overline{v})} \mathcal{G}'(V) \\ &= \mathcal{G}'_{\overline{y}}\end{aligned}$$
in the third equality the pair $(U, \overline{u})$ and the map $a : U \to X \times_Y V$ corresponds to the pair $(V, a \circ \overline{u})$. Since the stalk of $\mathcal{G}'$ (resp. $f_{spaces,étale}^{-1}\mathcal{G}'$) agrees with the stalk of $\mathcal{G}$ (resp. $f_{small}^{-1}\mathcal{G}$), see Equation (04JZ) the result follows. □